

Git Fetch & Comparison Learning Material

`git fetch origin dev`

Downloads the latest commits from remote dev without modifying your local branch.

`git log HEAD..origin/dev --oneline`

Shows commits that exist on remote dev but not in your current branch (HEAD).

`git log dev..origin/dev --oneline`

Shows commits that exist on remote dev but not in your local dev branch.

`git diff HEAD..origin/dev`

Shows line-by-line code differences between your current branch and remote dev.

`git diff dev..origin/dev`

Shows line-by-line differences between local dev and remote dev.

`git diff --stat HEAD..origin/dev`

Displays a summary of changed files and number of insertions/deletions.

`git diff --name-only HEAD..origin/dev`

Lists only the filenames that changed.

`git log --graph --oneline --decorate --all`

Visualizes the commit graph of all branches.

`git log --left-right --graph --oneline HEAD...origin/dev`

Shows commits unique to your current branch (<) and the remote branch (>).

`git show`

Displays the details of a specific commit.

HEAD vs dev

HEAD refers to the commit currently checked out. If you are on dev, then HEAD and dev are identical; otherwise they may differ.

Reading git log decorations

Example: commit abc123 (HEAD -> dev_, origin/dev_, fix_video_player)

HEAD -> dev_: current checked-out branch. origin/dev_: remote-tracking branch.

fix_video_player: another local branch pointing to the same commit. Multiple branches can point to the same commit because branches are lightweight pointers.